

Module Mini Project

Mile1 –Student Grade Calculation -- Project Design

Set of student data is available in an array of Student objects. If the given object has any errors, then, the program is expected to return appropriate error messages. Or, if given object has no data errors, then, we need to find the grade and print the same.

Package 1: **com.mile1.bean**

Class Student	
Variables	Description
String name; int marks []; String grade;	Instance variables
Constructors	
public Student () { }	To be Auto generated
public Student (String grade, int [] marks, String name) { // do the initialization }	To be Auto generated
Methods	
Provide public Getters and Setters for all instance variables	To be Auto generated

Package 2: **com. mile1.exception**

All the classes in this package should **extend** the Exception class.

Class	Method	Description
NullMarksArrayException	Override toString() method	Return "mark array is null"
NullNameException	Override toString() method	Return "name is null"
NullStudentObjectException	Override toString() method	Return "object is null"

Package 3: **com.mile1.service**

Class StudentReport	
Method	Description
<pre> public String findGrades (Student studentObject){ // write code here } </pre>	Assumption: Only valid objects are passed to this method. So, just concentrate on the logic part. Get the marks from the given object studentObject. if (any one of the marks is less than 35) then grade is "F"; else do the following: Find the Sum of all the marks. if (sum <150) then grade is "C"; else if (sum <200) then grade is "B"; else if (sum <250) then grade is "A"; else grade is "A+"; return the grade;
<pre> public String validate (Student s) throws NullPointerException, NullMarksArrayException, NullStudentObjectException { // write code here } </pre>	If Object passes as parameter itself is null, then, throw the NullStudentObjectException. Else do the following: Check whether there is any null data in the given object. We need to look for null inside the object. 1) If name is null then, throw the NullPointerException. 2) Else If marks array is null then throw the NullMarksException 3) Else If all data is valid, return "VALID".

Package3 com. mile1.service

Class StudentService	
Method	Description
<pre> public int findNumberOfNullMarksArray (Student s []) { // write code here } <i>If you are not careful, you will get NullPointerException in this method.</i> </pre>	This method is used to count the number of objects where the marks array is null. Let C=0; Check whether the s is not null. If so, then for all the objects in the s array: if the individual object is not null, then check whether the marks array is null. If so, increase C by 1. Return latest Count value;
<pre> public int findNumberOfNullName (Student s []) { // write code here } <i>Note: If you are not careful, you will get NullPointerException in this method.</i> </pre>	This method is used to count the number of objects where the name is null. // Code like above method // refer findNumberOfNullMarksArray
<pre> public int findNumberOfNullObjects (Student s []) { // write code here } <i>If you are not careful, you will get NullPointerException in this method.</i> </pre>	This method is used to count the number of null objects. Let C=0; Check whether the s is not null. If so, then for all the objects in the s array, if the individual object is null, then increase C by 1. Return latest Count value.

Package 4 com.mile1.main

Class StudentMain
Variables
static Student data [] = new Student [4];
Methods
<pre> StudentMain () { // Constructor for (int i = 0; i < data.length; i++) { data [i] = new Student (); } // initialize the objects data [0] = new Student ("Sekar", new int [] {85,75,95}); data [1] = new Student (null, new int [] {11,22,33}); data [2] = null; data [3] = new Student ("Manoj", null); } </pre>
<pre> public static void main (String [] args): // code as per the following explanation </pre> This main method is used to call the various methods defined in StudentReport class and StudentService class. 1) Call the validate method for all the objects available in data array. 2) If any exception arises during validation, catch the exception and call the printStackTrace () method on that object. 3) If validate method returns "VALID" then, call the findGrades method & print the result returned by findGrades method. Create StudentService Object. Using the above object, Call the findNumberOfNullMarksArray(data) method and print the result. Call the findNumberOfNullName(data) method and print the result. Call the findNumberOfNullObjects(data) method and print the result.

Sample Test Cases which are applied on the above project:

TC1 -- Calculate the grades for **valid** objects -- Check for A+ grade computation.

TC2 -- Calculate the grades for **valid** objects -- Check for F grade computation.

Test for validate method in service package:

Check whether the validate method handles the following situations.

TC3 -- If the Object is null, throw NullStudentObjectException ().

TC4-- If the Name is null, throw NullNameException ().

TC5 -- If the Marks array is null, throw NullMarksArrayException ().

Test for counting methods in service package:

TC6 -- Test findNumberOfNullName function.

TC7 -- Test findNumberOfNullObjects function.

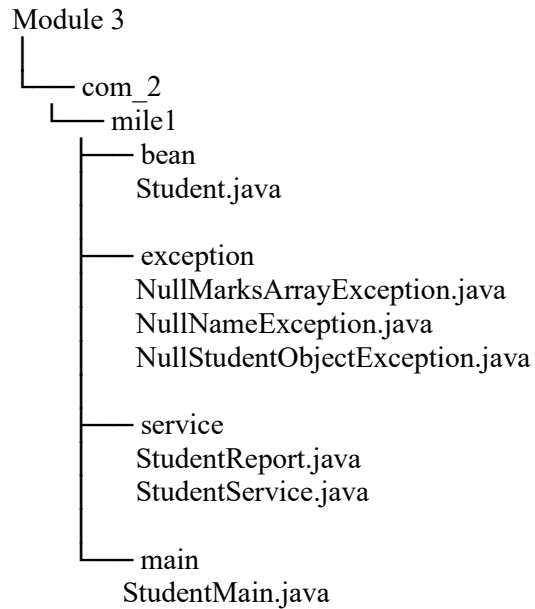
TC8 -- Test findNumberOfNullName

SAMPLE DATA SET1:

Note that your program will be tested with another set of data. Not this set of data.

```
public void init ()    {
    s [0] = new Student ("A1", new int [] {72,73,74});
    s [1] = new Student ("B1", new int [] {75,76,77});
    s [2] = new Student ("C1", new int [] {99,99,99});
    s [3] = new Student ("C3", new int [] {100,100,99});
    s [4] = new Student ("B2", new int [] {13,88,13});
    s [5] = new Student ("C3", new int [] {14,14,99});
    s [6] = new Student ("A2", new int [] {77,55,12});
    s [7] = new Student (null, new int [] {13,88,13});
    s [8] = new Student ("A2", null);
        // this is invalid object -- so no grade calculation
    s [9] = null; // this is invalid object -- so no grade calculation
    expectedgrades =new String [] {"A","A","A+","A+","F","F","F","F"};
}
```

Project Structure:



Code:

Student.java:

```
package com_2.mile1.bean;

public class Student {

    private String name;
    private int[] marks;
    private String grade;

    // Default Constructor
    public Student() {

    }

    // Parameterized Constructor
    public Student(String name, int[] marks) {
        this.name = name;
        this.marks = marks;
    }

    // Getter for Name
    public String getName() {
        return name;
    }

    // Setter for Name
    public void setName(String name) {
```

```
        this.name = name;
    }

    // Getter for Marks
    public int[] getMarks() {
        return marks;
    }

    // Setter for Marks
    public void setMarks(int[] marks) {
        this.marks = marks;
    }

    // Getter for Grade
    public String getGrade() {
        return grade;
    }

    // Setter for Grade
    public void setGrade(String grade) {
        this.grade = grade;
    }
}
```

NullMarksArrayException.java:

```
package com_2.mile1.exception;

public class NullMarksArrayException extends Exception {

    public NullMarksArrayException() {
        super();
    }

    public NullMarksArrayException(String message) {
        super(message);
    }
}
```

NullNameException.java:

```
package com_2.mile1.exception;

public class NullNameException extends Exception {

    public NullNameException() {
        super();
    }

    public NullNameException(String message) {
        super(message);
    }
}
```

```
}  
}
```

NullStudentObjectException.java:

```
package com_2.mile1.exception;  
  
public class NullStudentObjectException extends Exception {  
  
    public NullStudentObjectException() {  
        super();  
    }  
  
    public NullStudentObjectException(String message) {  
        super(message);  
    }  
  
}
```

StudentReport.java

```
package com_2.mile1.service;  
  
import com_2.mile1.bean.Student;  
import com_2.mile1.exception.NullMarksArrayException;  
import com_2.mile1.exception.NullNameException;  
import com_2.mile1.exception.NullStudentObjectException;  
  
public class StudentReport {  
  
    // Method to find the grade  
    public String findGrades(Student student) {  
  
        int sum = 0;  
  
        for (int mark : student.getMarks()) {  
  
            if (mark < 35) {  
                student.setGrade("F");  
                return "F";  
            }  
  
            sum += mark;  
        }  
  
        if (sum < 150) {  
            student.setGrade("C");  
            return "C";  
        } else if (sum < 200) {  
            student.setGrade("B");  
            return "B";  
        } else if (sum < 250) {
```



```
        student.setGrade("A");
        return "A";
    } else {
        student.setGrade("A+");
        return "A+";
    }
}

// Method to validate the student object
public String validate(Student student)
    throws NullStudentObjectException,
           NullNameException,
           NullMarksArrayException {

    if (student == null) {
        throw new NullStudentObjectException("Student object is null");
    }

    if (student.getName() == null) {
        throw new NullNameException("Student name is null");
    }

    if (student.getMarks() == null) {
        throw new NullMarksArrayException("Marks array is null");
    }

    return "VALID";
}
}
```

StudentService.java:

```
package com_2.mile1.service;

import com_2.mile1.bean.Student;

public class StudentService {

    // Count students whose marks array is null
    public int findNumberOfNullMarksArray(Student[] data) {

        int count = 0;

        for (Student student : data) {
            if (student != null && student.getMarks() == null) {
                count++;
            }
        }

        return count;
    }

    // Count students whose name is null
    public int findNumberOfNullName(Student[] data) {
```

```
        int count = 0;

        for (Student student : data) {
            if (student != null && student.getName() == null) {
                count++;
            }
        }

        return count;
    }

    // Count null student objects
    public int findNumberOfNullObjects(Student[] data) {

        int count = 0;

        for (Student student : data) {
            if (student == null) {
                count++;
            }
        }

        return count;
    }
}
```

Output:

```
▼ TERMINAL powershell + - ...
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\bean\Student.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\exception\NullMarksArrayException.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\exception\NullNameException.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\exception\NullStudentObjectException.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\service\StudentReport.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\service\StudentService.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> javac com_2\mile1\main\StudentMain.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 3> java com_2.mile1.main.StudentMain
Sekar Grade : A+
com_2.mile1.exception.NullNameException: Student name is null
    at com_2.mile1.service.StudentReport.validate(StudentReport.java:51)
    at com_2.mile1.main.StudentMain.main(StudentMain.java:27)
com_2.mile1.exception.NullStudentObjectException: Student object is null
    at com_2.mile1.service.StudentReport.validate(StudentReport.java:47)
    at com_2.mile1.main.StudentMain.main(StudentMain.java:27)
com_2.mile1.exception.NullMarksArrayException: Marks array is null
    at com_2.mile1.service.StudentReport.validate(StudentReport.java:55)
    at com_2.mile1.main.StudentMain.main(StudentMain.java:27)

Number of Null Marks Array : 1
Number of Null Names : 1
Number of Null Objects : 1
```